

Ratio and Proportion — Paper D

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Anything you cannot do in your head, do on paper.
- Give every ratio in its simplest form, and convert to a common unit before simplifying.
- The mark for each question is shown in square brackets on the right.
- Several questions do not give you the total. Work out what **one part** is worth first — everything else follows from it.
- Two questions on this paper have no obvious first step. If one stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – RATIOS THAT CHANGE

- 1.** The ratio of boys to girls in a club is $7 : 5$. Six more girls join the club, and the ratio becomes $7 : 6$. How many boys are in the club? [4]

Call one part k . The number of boys never changes, so write both ratios using the same k .

- 2.** Two numbers are in the ratio $3 : 5$. When 4 is added to each of them, the ratio becomes $2 : 3$. Find the two numbers. [4]

SECTION 2 – INVERSE PROPORTION

- 3.** 12 machines can complete an order in 15 days. All the machines work at the same rate.

(a) How long would 9 machines take to complete the same order? [2]

(b) The order must be finished in 10 days. How many machines are needed? [1]

(c) Explain in one sentence why your answers move in the opposite direction to the number of machines. [1]

4. A car travelling at 60 km/h completes a journey in 5 hours.

(a) How long would the same journey take at 75 km/h? [2]

(b) What speed would be needed to complete it in 3 hours? [1]

SECTION 3 – COMPARISONS AND CORRECTIONS

5. Three friends share the profit from a stall in the ratio of the hours they each worked. Asha worked 6 hours, Bhavna worked 9 hours and Chetan worked 12 hours. The profit was ₹4500.

(a) Work out how much each friend receives. [3]

- (b) Chetan says he should get half the profit because he worked the longest. Show [1]
by calculation that he is wrong.

6. Two shops sell the same juice.
Shop A: 1.5 litres for ₹99. **Shop B:** 2 litres for ₹126.

- (a) Work out the price per litre at each shop, and state which is better value. [2]

- (b) Shop A then offers “buy 2, get the third free”. Find Shop A’s new price per litre [2]
under this offer, and state which shop is now better value.

7. A recipe for 12 pancakes uses 200 g of flour, 300 ml of milk and 2 eggs. Ravi has [3]
500 g of flour, 700 ml of milk and 6 eggs. What is the greatest number of pancakes
he can make, and which ingredient runs out first?

Work out how many pancakes each ingredient on its own would allow, then take the smallest.

8. A student has written two answers in her book, and both are wrong.

[4]

(i) “The ratio of red to blue is 3 : 5, so $\frac{3}{5}$ of the counters are red.”

(ii) “A : B = 2 : 3 and B : C = 4 : 5, so A : B : C = 2 : 3 : 5.”

For each one, explain the mistake and give the correct answer.

End of paper. Check every answer has a unit or a form the question actually asked for.